

Plastic Waste Audit Report Generator for Institutions

Domain 10: Environment & Sustainability

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INTRODUCTION



- Increasing plastic waste poses significant environmental and sustainability challenges for institutions.
- Manual waste auditing and compliance assessment are time-consuming and error-prone.
- Institutions often struggle to comply with PWM Rules 2016 and EPR regulations.
- Our Solution automates waste analysis, compliance checking, and audit report generation using AI.
- The platform helps institutions reduce plastic waste and improve sustainability through actionable recommendations.

Problem Statement



Institutions such as universities and hospitals generate significant plastic waste without systematic tracking or reporting. Build a GenAI system that takes structured plastic waste audit data (type, quantity, disposal method) and generates a comprehensive plastic audit report aligned with India's Plastic Waste Management Rules 2016 and Extended Producer Responsibility guidelines.

Objectives

- Automate plastic waste data collection and analysis.
- Identify compliance gaps under PWM Rules 2016.
- Assess EPR obligations and sustainability performance.
- Generate AI-based recommendations and 12-month roadmaps.
- Produce official audit reports automatically.

Datasets Used



CPCB Plastic Waste Dataset

- State-wise plastic waste generation



National Plastic Waste Dataset

- Year-wise plastic waste trends



Polymer Usage Dataset

- Plastic usage by polymer type



Application Dataset

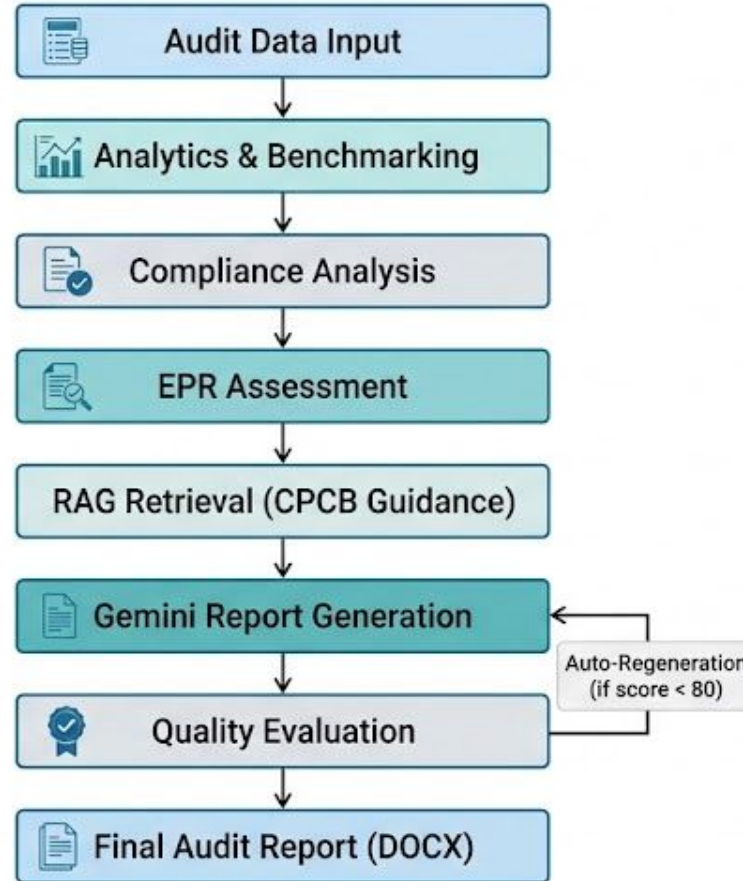
- Plastic usage by sector



CPCB EPR Guidance Manual

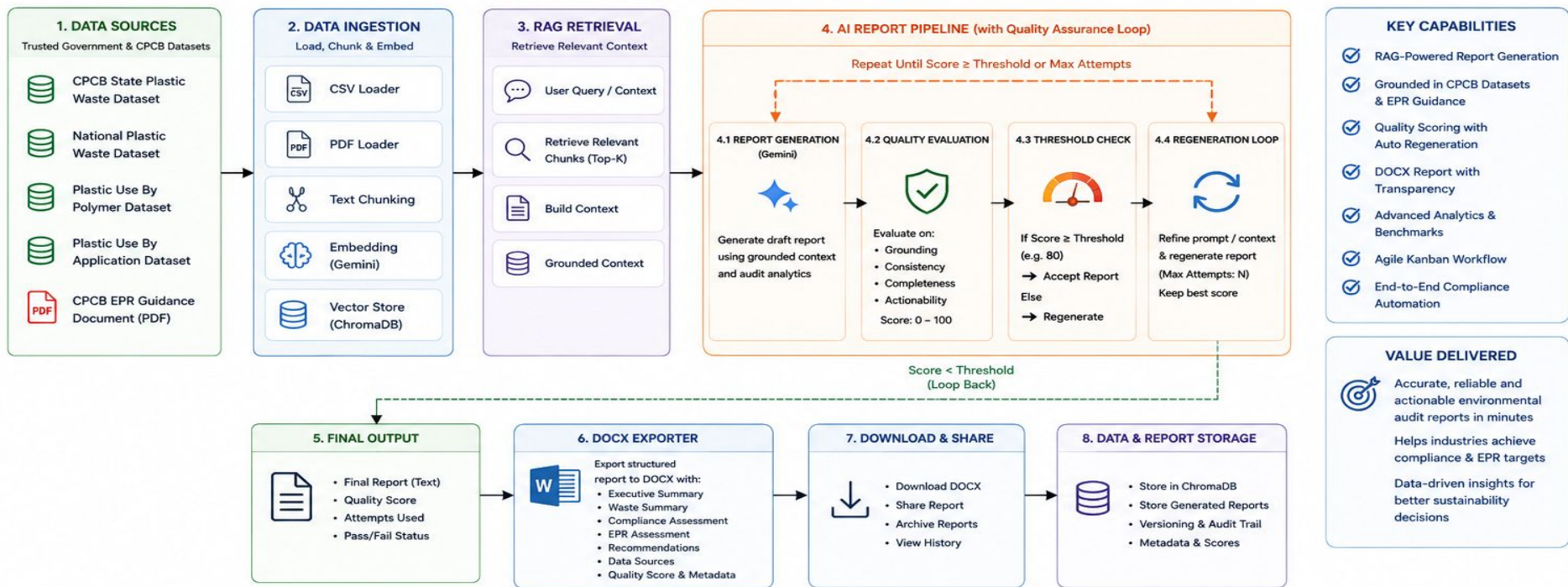
- Compliance knowledge base

Workflow of the System



PLASTIC WASTE AUDIT GENERATOR

AI-POWERED ENVIRONMENTAL COMPLIANCE PLATFORM



KEY CAPABILITIES

- ✓ RAG-Powered Report Generation
- ✓ Grounded in CPCB Datasets & EPR Guidance
- ✓ Quality Scoring with Auto Regeneration
- ✓ DOCX Report with Transparency
- ✓ Advanced Analytics & Benchmarks
- ✓ Agile Kanban Workflow
- ✓ End-to-End Compliance Automation

VALUE DELIVERED

- Accurate, reliable and actionable environmental audit reports in minutes
- Helps industries achieve compliance & EPR targets
- Data-driven insights for better sustainability decisions

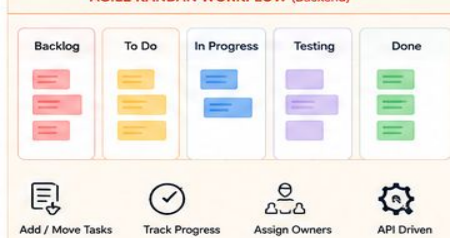
TECHNOLOGY FOUNDATION



SYSTEM INFRASTRUCTURE (Optional / Future Ready)



AGILE KANBAN WORKFLOW (Backend)



File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

Create Table Create Index Modify Table Delete Table Print Refresh

Name	Type	Schema
Tables (9)		
application_usage	CREATE TABLE	application_us
audits	CREATE TABLE	audits (id INT
institutions	CREATE TABLE	institutions (ic
national_trends	CREATE TABLE	national_trend
polymer_usage	CREATE TABLE	polymer_usage
reports	CREATE TABLE	reports (id IN
sqlite_sequence	CREATE TABLE	sqlite_sequenc
state_benchmarks	CREATE TABLE	state_benchma
waste_entries	CREATE TABLE	waste_entries
Indices (0)		
Views (0)		
Triggers (0)		

Edit Database Cell

Mode: Text

NULL

No cell active.

Type: NULL; Size: 0 bytes

Apply

Remote

Identity Select an identity to connect

Upload

DBHub.io

Local

Current Database

Name

Last modified

Size

SQL Log

Plot

DB Schema

Remote

UTF-8



Comparison: Generic LLMs vs Our System

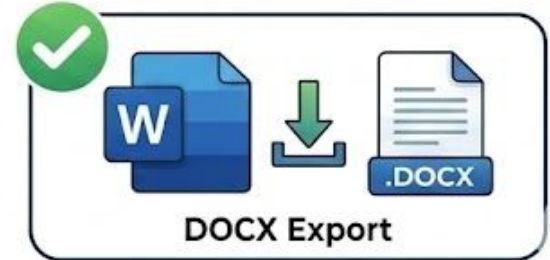
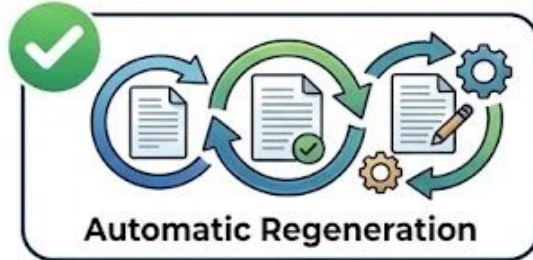
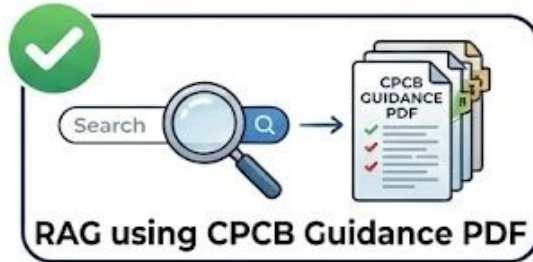
Feature	ChatGPT / Gemini (Generic LLMs)	Our System (Plastic Waste Audit Generator)
 Generate report from prompt	✓ Yes	✓ Yes
 Uses CPCB datasets	✗ No	✓ Yes
 Benchmark against state data	✗ No	✓ Yes
 Compliance with PWM Rules 2016	✗ No	✓ Yes
 EPR obligation assessment	✗ No	✓ Yes
 Uses CPCB EPR Guidance Document	✗ No	✓ Yes (via RAG)
 Quality evaluation & scoring	✗ No	✓ Yes (Auto scoring)
 Auto-regeneration (if score < threshold)	✗ No	✓ Yes
 Generate official DOCX report	✗ No	✓ Yes
 Explainable & actionable recommendations	✗ No	✓ Yes
 Audit-specific workflow	✗ No	✓ Yes (End-to-End platform)

USP-Quality Assurance Module



Metric	Weight	Description
Grounding	25%	Is the report supported by audit data, datasets, and CPCB guidance?
Consistency	25%	Are findings and recommendations logically consistent?
Completeness	25%	Are all required sections present?
Actionability	25%	Are recommendations practical and useful?

Features Implemented



Expected Output



Generated Report Contains

- Executive Summary
- Waste Analysis
- Compliance Assessment
- EPR Evaluation
- Recommendations
- 12-Month Roadmap
- Quality Score



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Creating Value
Leveraging Knowledge

Thank You